

AGENTIC AI WITH LANGGRAPH – SMART CAREER GUIDANCE AGENT

1. BRIEF INTRODUCTION OF THE PROJECT

The Smart Career Guidance Agent is an advanced Agentic AI system developed using LangGraph to help students and job seekers make informed career decisions. The system utilizes multiple AI agents that work together to analyze resumes, identify skill gaps, recommend suitable career paths, and generate personalized learning roadmaps. By leveraging Large Language Models (LLMs), the project provides intelligent, automated, and personalized career guidance. The system acts as a virtual career counselor that can assist users anytime and help them understand their strengths, weaknesses, and future career opportunities.

2. PROBLEM STATEMENT AND OVERVIEW

Many students and fresh graduates face difficulties in selecting the right career path because they lack awareness of industry requirements, emerging technologies, and job opportunities. Traditional career counseling methods are often expensive, time-consuming, and not tailored to individual needs. As a result, students may choose careers that do not match their skills or interests. The proposed Smart Career Guidance Agent addresses this issue by automatically evaluating user profiles, analyzing skills, identifying gaps, and recommending appropriate career options. Through the collaboration of multiple AI agents, the system delivers accurate career guidance and helps users make better career decisions.

3. TOOLS AND APPLICATIONS USED

Programming Language: Python

Frameworks and Libraries: LangGraph, LangChain, OpenAI API / Gemini API, FAISS Vector Database

Development Tools: VS Code, GitHub

Models Used: Large Language Models (LLMs), Resume Parsing Models, Embedding Models, Retrieval-Augmented Generation (RAG)

Applications: Career Guidance Systems, Resume Analysis Platforms, Skill Assessment Systems, Job Recommendation Platforms, and Learning Roadmap Generation Systems.

4. DETAILED DESCRIPTION OF THE SUB MODULES

The project consists of multiple AI agents. The Resume Analysis Agent extracts education details, technical skills, certifications, and project information from resumes. The Skill Gap Detection Agent compares the user's skills with industry requirements and identifies missing competencies. The Career Recommendation Agent suggests suitable job roles and career paths. The Learning Roadmap Agent creates a personalized learning plan to improve employability. The Job Matching Agent recommends relevant job opportunities based on the user's profile. Finally, the Report Generation Agent prepares a complete career guidance report containing recommendations and future growth strategies.

5. DESIGN AND FLOW OF THE PROJECT

Resume Upload → Resume Analysis Agent → Skill Gap Detection Agent → Career Recommendation Agent → Learning Roadmap Agent → Job Matching Agent → Final Career Guidance Report.

6. CONCLUSION / EXPECTED OUTPUT

The Smart Career Guidance Agent using LangGraph provides an intelligent and automated solution for career planning. The system helps users understand their strengths, identify skill gaps, explore suitable career opportunities, and follow a structured learning roadmap. The expected outputs include career recommendations, skill gap analysis, personalized learning plans, job role suggestions, and a comprehensive career guidance report. By integrating multiple AI agents, the project enhances decision-making, improves employability, and demonstrates the practical application of Agentic AI in solving real-world career development challenges.